

What Is Health Research?

Level 1 - Research Foundations | Student Learning Pack

How to use this pack

Read the teaching sections first, complete the worked example, do the short practice task, then take the basic knowledge check in RSRE. External references are optional further learning; you should be able to answer the core assessment from the Academy material.

Lesson 1: Start with the idea

Health research is a structured way of asking a question about health and collecting evidence that can help answer it. It is different from simply looking something up, following routine practice, or describing what happened once. Research begins with a clear problem and uses a planned method so another person can understand what was done and judge the evidence.

In health, research can describe a problem, identify risk factors, test an intervention, understand people's experiences, or improve how a health service works. A good research question is specific enough to answer with evidence. The question should guide the design, data collection, analysis and interpretation.

Example: "How common is self-medication with antibiotics among medical students at a university?" is a researchable question. "Antibiotic use is bad" is a statement, not a research question.

Remember: research does not begin with a result you want. It begins with a question you are willing to investigate honestly.

Lesson 2: Research or routine practice?

A clinic notices that many patients miss follow-up appointments. Staff begin calling patients one week before their next appointment as part of routine service improvement. If the team only implements and monitors the change for local improvement, that is primarily service practice. If the team develops a systematic study to answer a defined question, selects participants and variables in advance, analyzes the data, and intends to generate generalisable knowledge, it becomes research.

Ask yourself: 1. What question is being answered? 2. Is there a planned method? 3. What evidence will be collected? 4. Who needs to review or authorize the activity? 5. Is the goal local improvement, generalisable knowledge, or both?

There can be overlap between quality improvement and research, so researchers should follow the requirements of their institution and the responsible ethics or governance authority rather than deciding the classification informally.

Lesson 3: Why research matters in Rwanda

Research helps health workers move from assumptions to evidence. In Rwanda, a research question may come from malaria control, maternal health, non-communicable diseases, antimicrobial resistance, digital health, laboratory practice, health financing or the experience of students and patients.

A strong study can show the size of a problem, reveal who is affected, explain why a service is difficult to access, or test whether an intervention is useful. Research does not automatically change policy or practice; findings must also be interpreted, communicated and considered alongside feasibility, ethics and existing evidence.

A practical research mindset is: define the problem, ask an answerable question, choose a suitable method, collect trustworthy evidence, analyze it carefully, acknowledge uncertainty and communicate what the evidence actually supports.

Lesson 4: Worked example

Scenario: A university health centre wants to know whether students who receive a short malaria-prevention education session improve their knowledge.

Step 1 — Question: Does the education session improve malaria-prevention knowledge among students?

Step 2 — Outcome: a defined knowledge score before and after the session. Step 3 — Method: choose a design that can compare knowledge over time or between groups. Step 4 — Evidence: use a clear questionnaire and document how it was administered. Step 5 — Interpretation: compare the observed change without claiming more than the design can support.

Now create your own example from a health problem you have observed. Write the research question, the main outcome, who would provide the data, and what you would do next.

Lesson 5: Check your understanding

Before moving on, you should be able to explain in your own words:

- what health research is;
- how research differs from routine practice or casual information gathering;
- why a research question comes before choosing a method;
- why evidence must be collected and documented systematically; and
- why ethics and governance are part of responsible research.

If any of these points are unclear, review the lesson before taking the basic knowledge check. The assessment is designed to test understanding of the lesson, not advanced research expertise.

Competency check

After completing this module, you should be able to explain the core idea in your own words, identify a correct and incorrect application, and make a basic research decision using the concept.

RSRE note: The Academy is optional. Experienced researchers may use the diagnostic/competency pathway and enter other RSRE pillars directly.